Of course. Here are **clean, exam-ready notes** that are detailed enough for preparation but much easier to study and remember.

Suitable Visualization Techniques for Evaluating a Regression Model

Evaluating a regression model should not depend only on numerical measures such as R^2**, MAE, MSE, or RMSE**. Visualization techniques help us understand **how and where the model makes errors** and whether the assumptions of regression are satisfied.

The main assumptions checked using visualizations include:

- **Linearity** – the relationship between variables should be approximately linear.
- **Homoscedasticity** – the variance of errors should remain approximately constant.
- **Normality of residuals** – residuals should approximately follow a normal distribution.
- **Absence of influential outliers** – no single observation should excessively affect the model.

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1. Residual Plots

What is a Residual?

A **residual** is the difference between the actual value and the predicted value.

$$e_i = y_i - \hat{y}_i$$

Where:

- y_i= Actual value
- $\hat{y}_i$= Predicted value
- e_i= Residual (error)

A residual plot usually displays:

- **X-axis:** Predicted/Fitted values or an input feature
- **Y-axis:** Residuals

Interpretation of Residual Plots

Good Model

For a good regression model, residuals should be:

- Randomly scattered
- Distributed above and below zero
- Have approximately equal spread

Example:

Residual

|

+ | . . .

0 |---------- Predicted Value

- | . . . .

|

This indicates that the regression assumptions are reasonably satisfied.

Heteroscedasticity

If the spread of residuals changes as predicted values increase, it is called **heteroscedasticity**.

It often appears as a **funnel shape**.

Residual

|

| . .

| . . . .

0 |----------

| . .

| .

This means the error variance is **not constant**, which violates the homoscedasticity assumption.

Non-Linear Pattern

If residuals form a curved pattern, such as a U-shape, the model may be missing a non-linear relationship.

Residual

| . .

| . .

0 |------.---.---------

| .

This indicates **model misspecification**, meaning a linear regression model may not be suitable.

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2. Prediction Error Plot (Actual vs Predicted)

A prediction error plot compares:

- **Actual values** (y)
- **Predicted values** $(\hat{y})$

The plot includes a reference line:

$$y = x$$

This is a **45° diagonal line**, representing perfect prediction.

Interpretation

Points Close to the Diagonal

If points lie close to the diagonal:

$$\text{Actual Value} \approx \text{Predicted Value}$$

Therefore, predictions are accurate.

Points Far from the Diagonal

Points far from the diagonal represent large prediction errors.

Systematic Deviation

If points consistently appear above or below the line in a particular region, the model may have systematic bias.

For example:

- High actual values consistently below the line may indicate poor prediction at high values.

- The model may systematically under-predict or over-predict certain ranges.

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3. Normal Q-Q Plot

A **Normal Quantile-Quantile (Q-Q) Plot** is used to check whether residuals follow a normal distribution.

It compares:

- The actual distribution of residuals
- The theoretical normal distribution

Interpretation

Normally Distributed Residuals

If residuals are approximately normally distributed, points lie close to the reference line.

Theoretical Quantiles

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/

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Non-Normal Residuals

Large deviations from the line indicate that residuals may not follow a normal distribution.

Common patterns include:

- Curves
- S-shaped patterns
- Large deviations at the ends (tails)

These may indicate:

- Outliers
- Heavy-tailed distributions
- Skewness
- Violation of the normality assumption

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4. Cook's Distance Plot

Cook's Distance is used to identify **influential observations**.

An observation is influential if removing it significantly changes the regression model.

Cook's Distance measures how much the model's fitted coefficients change when a particular observation is removed.

Influence Threshold

A commonly used threshold is:

$$\frac{4}{n}$$

Where:

- n= Number of observations

Any observation with Cook's Distance greater than this threshold should be investigated.

Interpretation

Small Cook's Distance

A small value indicates that removing the observation does not significantly affect the model.

Large Cook's Distance

A large value indicates that the observation has a strong influence on the regression model.

Such observations may be:

- Data entry errors
- Unusual cases
- Extreme values
- High-leverage observations

They should be investigated before removing them.

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5. Regularization Alpha Selection Plots

These plots are mainly used with:

- **Ridge Regression**
- **Lasso Regression**

Both methods use a regularization parameter called:

$$\alpha$$

The value of αcontrols the amount of penalty applied to model coefficients.

A. Validation Error Plot

A validation error plot shows how prediction error changes as αchanges.

It helps select the optimal value of α.

The best value is usually the one that produces the **lowest validation or cross-validation error**.

This helps achieve a good balance between:

- **Bias**
- **Variance**

B. Coefficient Path Plot

A coefficient path plot shows how regression coefficients change as αincreases.

As αincreases:

- Coefficients shrink toward zero.

Ridge Regression

Coefficients become smaller but generally do not become exactly zero.

Lasso Regression

Some coefficients may become exactly:

$$0$$

This automatically performs **feature selection**.

Therefore, Lasso helps identify features that are less important.

Illustrative Example: House Price Prediction

Suppose we build a linear regression model to predict **House Price** based on **Built-up Area**.

The fitted regression equation is:

$$\hat{Y} = 2.484 + 0.0495X$$

Where:

- X= Area in square feet
- $\hat{Y}$= Predicted house price in ₹ Lakhs

Sample Data

House	Area (sq. ft.)	Actual Price	Predicted Price	Residual
1	500	25.0	27.234	-2.234
2	600	32.0	32.184	-0.184
3	700	36.0	37.134	-1.134
4	800	41.0	42.084	-1.084
5	900	47.0	47.034	-0.034
6	1000	52.0	51.984	0.016
7	1100	58.0	56.934	1.066
8	1200	62.0	61.884	0.116
9	1300	65.0	66.834	-1.834
10	1400	102.0	110.0	-8.0

House 10 represents an unusual or extreme observation.

Diagnostic Analysis of the Model

A. Prediction Error Plot

When actual prices are plotted against predicted prices:

- Houses 1–9 lie close to the $45°$ diagonal line.
- This indicates good prediction accuracy for normal houses.
- House 10 lies far from the diagonal line.

Therefore, House 10 has a large prediction error.

Conclusion

The model performs well for most houses but performs poorly for House 10.

B. Residuals vs Fitted Values Plot

For Houses 1–9:

- Residuals are scattered around zero.
- No major curve is visible.
- No clear funnel shape is visible.

Therefore:

- The linearity assumption appears reasonable.
- The homoscedasticity assumption appears reasonably satisfied.

However, House 10 has a very large negative residual.

This suggests an unusual observation that requires investigation.

C. Normal Q-Q Plot

For most observations:

- Residuals lie approximately close to the reference line.

However:

- House 10 deviates strongly from the line.

This suggests that the residual distribution may not be perfectly normal.

The extreme residual may indicate:

- An outlier
- A data error
- An unusual house property

D. Cook's Distance Plot

For $n = 10$observations:

$$\text{Threshold} = \frac{4}{n}$$
$$= \frac{4}{10}$$
$$= 0.40$$

Suppose Houses 1–9 have Cook's Distance values below 0.03.

These values are well below the threshold of 0.40.

Therefore, these observations are not highly influential.

However, House 10 has:

$$D_{10} = 3.830$$

Since:

$$3.830 > 0.40$$

House 10 is considered a **highly influential observation**.

High Leverage and Influential Outliers

House 10 also has high leverage.

Leverage measures how unusual an observation's input value is compared with other observations.

An observation with:

- High leverage
- Large residual
- High Cook's Distance

can strongly influence the regression model.

Therefore, House 10 should be carefully investigated.

Possible reasons include:

- Data entry error
- Measurement error

- An unusual property
- Missing important features
- A genuine but extreme observation

Summary Table

Visualization	Purpose	Good Pattern	Problem Indicated
Residual Plot	Checks linearity and constant variance	Random scatter around zero	Curve or funnel pattern
Actual vs Predicted Plot	Checks prediction accuracy	Points close to $y = x$line	Points far from line
Normal Q-Q Plot	Checks normality of residuals	Points close to reference line	Curves or large deviations
Cook's Distance Plot	Detects influential observations	Small Cook's Distance values	Values above $4/n$
Alpha Selection Plot	Selects optimal regularization strength	Minimum validation error	Too much/too little regularization
Coefficient Path Plot	Shows coefficient shrinkage	Stable important coefficients	Important coefficients shrinking excessively

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Final Exam Conclusion

Visualization techniques are essential for evaluating regression models because numerical metrics alone cannot reveal all problems in a model. **Residual plots** help check linearity and homoscedasticity, **prediction error plots** show prediction accuracy, **Q-Q plots** check the normality of residuals, and **Cook's Distance plots** identify influential outliers. For Ridge and Lasso regression, **regularization plots** help select the optimal value of $\alpha$and understand coefficient shrinkage. Therefore, visual diagnostics provide a more complete understanding of regression model performance and help identify problems before deploying the model.

Quick memory trick

R-P-Q-C-A

- **R** → Residual Plot

- **P** → Prediction Error Plot
- **Q** → Q-Q Plot
- **C** → Cook's Distance
- **A** → Alpha/Regularization Plot

This covers the entire topic in a logical exam-friendly sequence.